

Math 237B Final

Zih-Yu Hsieh

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1 D

Problem 1. Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be morphisms of schemes such that $g \circ f$ is proper, g is quasi-projective and f is surjective. Prove or disprove that g is proper.

Solution. We'll prove that g is proper. We'll prove relevant statements in the following order:

- (a) g is separated, and f is proper.
- (b) g is of finite type.
- (c) g is universally closed.

Finally, use these results we can conclude that g is proper.

Proof of (a):

Since $g : Y \rightarrow Z$ is quasi-projective, there exists some scheme Y' , together with open immersion $\iota : Y \hookrightarrow Y'$ and projective morphism $\pi : Y' \rightarrow Z$, such that the following commutative diagram holds:

$$\begin{array}{ccc} Y & \xrightarrow{\iota} & Y' \\ g \downarrow & \swarrow \pi & \\ Z & & \end{array}$$

Which, recall that any open immersion is separated, and projective morphisms are proper (in particular separated), so $g = \pi \circ \iota$ is also separated.

As a consequence, since $g \circ f$ is proper, while g is separated, we concluded that f is proper also.

Proof of (b):

First, let's show g is locally of finite type: Using the diagram in part (a), since $\pi : Y' \rightarrow Z$ is projective, it is proper (hence of finite type), so for any open affine subset $U \subseteq Z$ (say $U \cong \text{Spec}(A)$), such that $\pi^{-1}(U) = \bigcup_{i=1}^n V_i$, where each V_i is affine open subset, and $V_i \cong \text{Spec}(B_i)$ for some B_i that's finitely generated A -algebra.

Then, with open immersion $\iota : Y \hookrightarrow Y'$, one has $\iota^{-1}(\pi^{-1}(U)) = Y \cap \pi^{-1}(U) = \bigcup_{i=1}^n Y \cap V_i$. Since Y is open in Y' , each $Y \cap V_i$ is an open subset of V_i under its subspace topology, hence each $Y \cap V_i = \bigcup_{j \in J_i} D(f_{ij})$ for some $f_{ij} \in B_i$ (running over some index set J_i). Because each $D(f_{ij}) \cong \text{Spec}\left((B_i)_{f_{ij}}\right)$ (which $(B_i)_{f_{ij}}$ is a finitely generated B_i -algebra), then with B_i being a finitely generated A -algebra, $(B_i)_{f_{ij}}$ is a finitely generated A -algebra also.

As a result, one has $g^{-1}(U) = \iota^{-1}(\pi^{-1}(U)) = \bigcup_{i=1}^n Y \cap V_i = \bigcup_{i=1}^n \bigcup_{j \in J_i} D(f_{ij})$, where each $D(f_{ij}) \cong \text{Spec}\left((B_i)_{f_{ij}}\right)$ has $(B_i)_{f_{ij}}$ being a finitely generated A -algebra. Since $U \subseteq Z$ is arbitrary affine open subset, this shows g is locally of finite type.

Then, let's show g is of finite type: Use the previous notation, since $g \circ f$ is finite type, it's quasi-compact, hence with affine open subset $U \subseteq Z$, one has $(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U))$ being quasi-compact. However, since f is surjective, one has $g^{-1}(U) = f(f^{-1}(g^{-1}(U)))$. Because it's an image of a quasi-compact set under continuous map, $g^{-1}(U)$ is quasi-compact. Hence, the previous cover for locally of finite type property, say $g^{-1}(U) = \bigcup_{i=1}^n \bigcup_{j \in J_i} D(f_{ij})$, can be chosen as finite. This shows that g is of finite type.

Proof of (c):

To prove g is universally closed, for any morphism of schemes $h : Z' \rightarrow Z$ consider the following commutative diagram of multiple fibre products:

$$\begin{array}{ccc}
 (Z' \times_Z Y) \times_Y X & \xrightarrow{h''} & X \\
 f' \downarrow & & \downarrow f \\
 Z' \times_Z Y & \xrightarrow{h'} & Y \\
 g' \downarrow & & \downarrow g \\
 Z' & \xrightarrow{h} & Z
 \end{array}$$

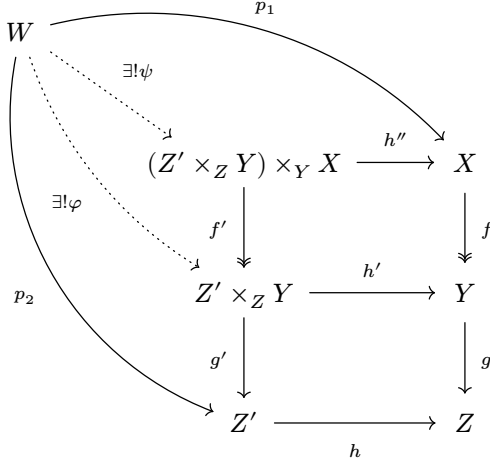
Which:

- $g' : Z' \times_Z Y \rightarrow Z'$ denotes the base change of $g : Y \rightarrow Z$ by $h : Z' \rightarrow Z$.
- $f' : (Z' \times_Z Y) \times_Y X \rightarrow Z' \times_Z Y$ denotes the base change of $f : X \rightarrow Y$ by $h' : Z' \times_Z Y \rightarrow Y$ (the pullback of $h : Z' \rightarrow Z$), which the surjectivity of f ensures the surjectivity of f' . And, since part (a) shows f is proper (which is universally closed), then f' is closed.

Now, we ought to show $(Z' \times_Z Y) \times_Y X$ together with $g' \circ f'$ and h'' satisfies the universal property of the fibre product $Z' \times_Z X$. For any scheme W together with two morphisms $p_1 : W \rightarrow X$ and $p_2 : W \rightarrow Z'$, such that $(g \circ f) \circ p_1 = h \circ p_2$. Then, the universal property of $Z' \times_Z Y$ guarantees a unique morphism of schemes $\varphi : W \rightarrow Z' \times_Z Y$, such that $p_2 = g' \circ \varphi$, and $f \circ p_1 = h' \circ \varphi$, as follow:

$$\begin{array}{ccc}
 W & \xrightarrow{p_1} & X \\
 \exists! \varphi \swarrow & & \downarrow f \\
 (Z' \times_Z Y) \times_Y X & \xrightarrow{h''} & X \\
 f' \downarrow & & \downarrow f \\
 Z' \times_Z Y & \xrightarrow{h'} & Y \\
 g' \downarrow & & \downarrow g \\
 Z' & \xrightarrow{h} & Z
 \end{array}$$

Furthermore, with the universal property of $(Z' \times_Z Y) \times_Y X$, the equality $f \circ p_1 = h' \circ \varphi$ guarantees a unique morphism of schemes $\psi : W \rightarrow (Z' \times_Z Y) \times_Y X$, such that $\varphi = f' \circ \psi$ and $p_1 = h'' \circ \psi$, as follow:



This proves both the existence and the uniqueness of the morphism $\psi : W \rightarrow (Z' \times_Z Y) \times_Y X$, such that $p_2 = g' \circ \varphi = (g' \circ f') \circ \psi$, and $p_1 = h'' \circ \psi$ (as any $\psi' : W \rightarrow (Z' \times_Z Y) \times_Y X$ with such property is enforced to be equal to ψ , based on individual fibre product's property). So, $(Z' \times_Z Y) \times_Y X$ indeed satisfies the universal property of $Z' \times_Z X$, showing that $g' \circ f'$ is the base change of the morphism $(g \circ f) : X \rightarrow Z$. Which, since $g \circ f$ is proper, $g' \circ f'$ is closed.

Finally, with f' and $g' \circ f'$ being closed, and f' being surjective, we claim that g' is closed:

For any closed subset $V \subseteq Z' \times_Z Y$, by the continuity of f' , the subset $(f')^{-1}(V) \subseteq (Z' \times_Z Y) \times_Y X$ is closed. Hence, $(g' \circ f')((f')^{-1}(V)) \subseteq Z'$ is closed (by closeness of $g' \circ f'$).

Also, since f' is surjective, one simply has $f'((f')^{-1}(V)) = V$. Hence, the above set equality shows that $(g' \circ f')((f')^{-1}(V)) = g'(V) \subseteq Z'$ is closed. This proves that g' is closed.

With g' being based change of g over arbitrary morphism of schemes $h : Z' \rightarrow Z$, this shows that g is universally closed.

With all information above, we can conclude that g is separated, finite type, and universally closed, hence proper.

2 ND (finish the segre embedding)

Problem 2. Let $f : X \rightarrow S$ and $g : Y \rightarrow S$ be projective morphisms. Prove or disprove that the projection map $X \times_S Y \rightarrow S$ is projective.

Solution. We'll show that $X \times_S Y \rightarrow S$ is projective. We'll prove the following two statements:

- (a) Base change of projective morphisms are projective.
- (b) Composition of projective morphisms are projective.

Then, the two will help conclude the desired result.

Proof of (a):

First, we'll show that base change preserves projective spaces: More precisely, given any $n \in \mathbb{N}$, with morphisms $g : Y \rightarrow S$, the projective n -space over S , $\pi : \mathbb{P}_S^n \rightarrow S$ has fibre product $Y \times_S \mathbb{P}_S^n$ satisfies the universal property of $\mathbb{P}_Y^n$.

Indeed, with the definition of $\mathbb{P}_S^n := S \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n$, we form the following commutative diagram:

$$\begin{array}{ccccc}
 Y \times_S \mathbb{P}_S^n & \xrightarrow{g'} & \mathbb{P}_S^n & \xrightarrow{s'} & \mathbb{P}_{\mathbb{Z}}^n \\
 \pi' \downarrow & & \downarrow \pi & & \downarrow \varphi \\
 Y & \xrightarrow{g} & S & \xrightarrow{s} & \text{Spec}(\mathbb{Z})
 \end{array}$$

Where, both small squares are fibre product diagrams. Also, the morphism $s \circ g : Y \rightarrow \text{Spec}(\mathbb{Z})$ is the unique morphism based on the terminal property of $\text{Spec}(\mathbb{Z})$ in the category of schemes.

Now, given any scheme W together with two morphisms $p_1 : W \rightarrow \mathbb{P}_{\mathbb{Z}}^n$ and $p_2 : W \rightarrow Y$, such that $\varphi \circ p_1 = (s \circ g) \circ p_2$, then the universal property of $\mathbb{P}_S^n$ generates a unique morphism $\psi : W \rightarrow \mathbb{P}_S^n$, such that $p_1 = s' \circ \psi$, and $g \circ p_2 = \pi \circ \psi$, as follow:

$$\begin{array}{c}
 \begin{array}{c}
 W \\
 \begin{array}{c}
 \xrightarrow{p_1} \mathbb{P}_{\mathbb{Z}}^n \\
 \text{---} \exists! \psi \text{---} \\
 \xrightarrow{p_2} Y
 \end{array}
 \end{array}
 \begin{array}{ccccc}
 Y \times_S \mathbb{P}_S^n & \xrightarrow{g'} & \mathbb{P}_S^n & \xrightarrow{s'} & \mathbb{P}_{\mathbb{Z}}^n \\
 \pi' \downarrow & & \downarrow \pi & & \downarrow \varphi \\
 Y & \xrightarrow{g} & S & \xrightarrow{s} & \text{Spec}(\mathbb{Z})
 \end{array}
 \end{array}$$

Then, with the universal property of $Y \times_S \mathbb{P}_S^n$, the second equality $g \circ p_2 = \pi \circ \psi$ guarantees a unique morphism $\rho : W \rightarrow Y \times_S \mathbb{P}_S^n$, such that $\psi = g' \circ \rho$, and $p_2 = \pi' \circ \rho$, as follow:

$$\begin{array}{c}
 \begin{array}{c}
 W \\
 \begin{array}{c}
 \xrightarrow{p_1} \mathbb{P}_{\mathbb{Z}}^n \\
 \text{---} \exists! \psi \text{---} \\
 \text{---} \exists! \rho \text{---} \\
 \xrightarrow{p_2} Y
 \end{array}
 \end{array}
 \begin{array}{ccccc}
 Y \times_S \mathbb{P}_S^n & \xrightarrow{g'} & \mathbb{P}_S^n & \xrightarrow{s'} & \mathbb{P}_{\mathbb{Z}}^n \\
 \pi' \downarrow & & \downarrow \pi & & \downarrow \varphi \\
 Y & \xrightarrow{g} & S & \xrightarrow{s} & \text{Spec}(\mathbb{Z})
 \end{array}
 \end{array}$$

Again, this proves both the existence and the uniqueness of the morphism $\rho : W \rightarrow Y \times_S \mathbb{P}_S^n$, such that $p_1 = s' \circ \psi = (s' \circ g') \circ \rho$, and $p_2 = \pi' \circ \rho$. Hence, $Y \times_S \mathbb{P}_S^n$ satisfies the universal property of $\mathbb{P}_Y^n := Y \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n$, showing $Y \times_S \mathbb{P}_S^n \cong \mathbb{P}_Y^n$.

Now, we'll prove that base change preserves projective morphisms, again by the universality argument: Given a projective morphism $f : X \rightarrow S$, and another morphism $g : Y \rightarrow S$, we know there exists a suitable $n \in \mathbb{N}$, such that there exists a closed immersion $\iota : X \hookrightarrow \mathbb{P}_S^n$ and the projection $\pi : \mathbb{P}_S^n \rightarrow S$, such that $f = \pi \circ \iota$.

Which, based on the previous argument, we'll simply denote $Y \times_S \mathbb{P}_S^n$ as $\mathbb{P}_Y^n$. Together with the previous diagrams, we have the following:

$$\begin{array}{ccccc}
 X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n & \xrightarrow{g''} & X & & \\
 \downarrow \iota' & & \downarrow \iota & & \\
 \mathbb{P}_Y^n & \xrightarrow{g'} & \mathbb{P}_S^n & \xrightarrow{s'} & \mathbb{P}_{\mathbb{Z}}^n \\
 \downarrow \pi' & & \downarrow \pi & & \downarrow \varphi \\
 Y & \xrightarrow{g} & S & \xrightarrow{s} & \text{Spec}(\mathbb{Z})
 \end{array}$$

Which, we took the fibre product $X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n$, and the morphism $\iota' : X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n \hookrightarrow \mathbb{P}_Y^n$ is the pullback of the closed immersion $\iota : X \hookrightarrow \mathbb{P}_S^n$, hence ι' is also a closed immersion.

Now, we claim that it satisfies the universal property of $X \times_S Y$: In general, the above proof of preserving projective spaces shows that successive fibre product preserves fibre product in the full generality. By modifying the labels / objects a bit, we indeed get a proof that $X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n$ satisfies the universal property of $X \times_S Y$ as fibre product, hence $X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n \cong X \times_S Y$.

Finally, by denoting $X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n$ as $X \times_S Y$, the above diagram reduces to the following fibre product diagram:

$$\begin{array}{ccc}
 X \times_S Y & \xrightarrow{g''} & X \\
 \downarrow \iota' & & \downarrow \iota \\
 \mathbb{P}_Y^n & & \mathbb{P}_S^n \\
 \downarrow \pi' & & \downarrow \pi \\
 Y & \xrightarrow{g} & S
 \end{array}
 \quad \begin{array}{c} \curvearrowright f \end{array}$$

Which, the pullback of f is $\pi' \circ \iota' : X \times_S Y \rightarrow Y$, which is precisely a closed immersion into $\mathbb{P}_Y^n$, then a projection onto Y . This shows that the pullback of f is still projective, hence projective morphisms are preserved under base change.

Proof of (b):

3 D

Problem 3. Let $f : X \rightarrow Y$ be a morphism of separated scheme of finite type over a Noetherian scheme S . Let $Z \subset X$ be a closed subscheme which is proper over S . Show that $f(Z)$ is closed in Y .

Solution. To show $f(Z) \subseteq Y$ is closed, we'll show that with the closed immersion $\iota : Z \hookrightarrow X$, one has $f \circ \iota : Z \rightarrow Y$ is proper, which will show $f \circ \iota(Z) = f(Z)$ is closed in Y .

First, let $\varphi_X : X \rightarrow S$ and $\varphi_Y : Y \rightarrow S$ be the morphism that define X, Y as separated schemes of finite type over S , this implies both φ_X, φ_Y are of finite type and separated morphisms.

Also, consider the closed immersion $\iota : Z \hookrightarrow X$, since Z is proper over S , we have this morphism $\varphi_X \circ \iota : Z \rightarrow S$ be proper.

Now, since $f : X \rightarrow Y$ is a morphism of schemes over S , we have the following commutative diagram:

$$\begin{array}{ccccc}
 & & Z & & \\
 & & \searrow & & \\
 & & \iota & & \\
 & & \searrow & & \\
 & X & \xrightarrow{f} & Y & \\
 & \searrow & & \swarrow & \\
 & \varphi_X & & \varphi_Y & \\
 & & S & &
 \end{array}$$

Or, $\varphi_X = \varphi_Y \circ f$, which implies $\varphi_X \circ \iota = \varphi_Y \circ (f \circ \iota)$. Since $\varphi_X \circ \iota$ is proper, with φ_Y being separated, we have $f \circ \iota$ being proper also (which is a closed map). With Z being closed in itself, $f \circ \iota(Z) = f(Z) \subseteq Y$ is closed. This finishes our proof.

4 ND (Type some ideal up, fxxk you localization)

Problem 4. Let k be a field and let $R = k[X, Y, Z]/I$, where $I = (X - YZ, XZ - Y^2)$. Let $W = \text{Spec}(R)$.

- (a) Is W a reduced scheme? Justify your answer.
 - (b) Is W irreducible? If not, what are its irreducible components?
 - (c) Prove or disprove that W has infinitely many rational points if k is infinite (recall that a point $x \in W$ is called rational point if the canonical map $k \rightarrow k(x)$ is an isomorphism, where $k(x)$ is the residue field of the local ring $\mathcal{O}_{X,x}$).
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Solution.

- (a) This is kind of fucked (look at localization at (X, Y) , I believe this causes Y to be zero divisor, as we have $X = YZ = \frac{Y^2}{Z}$, so $Y^2 = YZ^2$, or $Y(Y - Z^2) = 0$).
- (b) Also kind of fucked (classically the solutions are $(X, Y, Z) = (t^3, t^2, t)$, corresponding to ideal $(X - Z^3, Y - Z^2)$ prime, refer to 237A HW about twisted cubics; the other one is $(X, Y, Z) = (0, 0, t)$, corresponding to ideal (X, Y) also prime). Hence, a natural guess here is $V((X - Z^3, Y - Z^2))$ and $V((X, Y))$ should cover up the space (while both irreducible components). Here, I guess we can use prime avoidance theorem / prime inclusion theorem in a suitable way.
- (c) If k is infinite, then consider the stalks of $(x, y, z - t)$ for all $t \in k$, and/or $(x - t^3, y - t^2, z - t)$, I believe these two do the job (in particular, because they're maximal ideals, which correspond to closed point in all ring). Here, I believe it uses the fact that localization preserves surjectivity, and the fact that they're local homomorphisms (so by taking infinite maximal ideals containing I , the quotient should form an isomorphism, bc. the residue field of a maximal ideal containing I should be k).

5 ND

Problem 5. Let $f : X \rightarrow Y$ be a surjective morphism between two schemes which are finite type and separated integral schemes over a field. Assume that X is an affine scheme. Prove or disprove that Y is an affine scheme if and only if it is a quasi-affine scheme (recall that quasi-affine means open subscheme of an affine scheme).

Solution. It's clear that affine $\implies$ quasi-affine (regardless of the situation). So, the converse is the actual nontrivial part to prove / disprove